ModE Project 5

June 2026

1 MILP Formulation

Objective

$$\min_{\mathbf{x}} \quad c_{\mathrm{G}} \sum_k E_{\mathrm{Gas},k} + c_{\mathrm{el}} \sum_k E_{\mathrm{el},k}$$

with

$$\begin{aligned} E_{\mathrm{el},k} &= \Delta t \cdot P_{\mathrm{grid},k} \\ E_{\mathrm{Gas},k} &= \Delta t \left[\sum_{i \in \mathcal{B}} \dot{Q}^{\mathrm{in}}_{\mathrm{B},i,k} + \sum_{i \in \mathcal{CHP}} \dot{Q}^{\mathrm{in}}_{\mathrm{CHP},i,k} \right] \end{aligned}$$

Decision variables At each time step $k \in \{1, \ldots, N\}$, the decision variable vector of the independent variables is

$$\tilde{\mathbf{x}}_k = \begin{matrix} \Big[\dot{Q}^{\mathrm{in}}_{\mathrm{B},1,k},\ \dot{Q}^{\mathrm{in}}_{\mathrm{B},2,k},\ \dot{Q}^{\mathrm{in}}_{\mathrm{CHP},1,k},\ \dot{Q}^{\mathrm{in}}_{\mathrm{CHP},2,k},\ \dot{Q}^{\mathrm{in}}_{\mathrm{TES},k},\ \dot{Q}^{\mathrm{out}}_{\mathrm{TES},k}, \\ P_{\mathrm{grid},k},\ \delta_{\mathrm{B},1,k},\ \delta_{\mathrm{B},2,k},\ \delta_{\mathrm{CHP},1,k},\ \delta_{\mathrm{CHP},2,k},\ \delta^{\mathrm{in}}_{\mathrm{TES},k},\ \delta^{\mathrm{out}}_{\mathrm{TES},k} \Big]^\top \end{matrix}$$

With all variables being continuous and in $\mathbb{R}^n$ except the binary variables $\delta \in \{0, 1\}$.

The auxiliary variables that could eliminated by substitution (to get a dense formulation) are at each time step $k \in \{1, \ldots, N\}$

$$\mathbf{y}_k = \begin{matrix} \Big[E_{\mathrm{TES},k},\ \dot{Q}^{\mathrm{out}}_{\mathrm{B},1,k},\ \dot{Q}^{\mathrm{out}}_{\mathrm{B},2,k},\ \dot{Q}^{\mathrm{out}}_{\mathrm{CHP},1,k},\ \dot{Q}^{\mathrm{out}}_{\mathrm{CHP},2,k}, \\ P^{\mathrm{out}}_{\mathrm{CHP},1,k},\ P^{\mathrm{out}}_{\mathrm{CHP},2,k},\ E_{\mathrm{G},k},\ E_{\mathrm{el},k} \Big]^\top \end{matrix}$$

They are kept as decision variables for the sparse formulation. The full decision variable vector becomes

$$\mathbf{x}_k = \begin{bmatrix} \tilde{\mathbf{x}}_k^\top & \mathbf{y}_k^\top \end{bmatrix}^\top .$$

Time discretization The continuous time horizon is discretized into N uniform time steps indexed by $k \in \{0, 1, \ldots, N\}$, with

$$t \in [0, t_f], \qquad t_k = k \cdot \Delta t, \qquad \Delta t = t_{k+1} - t_k \quad \forall k, \qquad t_f = N \, \Delta t$$

Demand as well as outputs of conversion components are assumed constant throughout each time period. The differential equation **TES**(I) is discretized exactly using the matrix exponential. This yields the discrete heat contents of the TES $E_{\mathrm{TES},k}$.

Subject to:

TES

$$\begin{aligned}
&\text{(I)} && \dot{E}_{\mathrm{TES}}(t) = \eta^{\mathrm{in}}_{\mathrm{TES}} \, \dot{Q}^{\mathrm{in}}_{\mathrm{TES}}(t) - \frac{\dot{Q}^{\mathrm{out}}_{\mathrm{TES}}(t)}{\eta^{\mathrm{out}}_{\mathrm{TES}}} - \frac{E_{\mathrm{TES}}(t)}{\tau^{\mathrm{loss}}} \\
&\text{discretized, see 1.1:} && \left(E_{\mathrm{TES},k+1} = a \, E_{\mathrm{TES}_k} + b_1 \, \dot{Q}^{\mathrm{in}}_{\mathrm{TES},k} + b_2 \, \dot{Q}^{\mathrm{out}}_{\mathrm{TES},k} \right) \\
&\text{(II)} && E_{\mathrm{TES},k} \leq E^{\mathrm{nom}}_{\mathrm{TES}} \\
&\text{(III)} && -E_{\mathrm{TES},k} \leq -E^{\mathrm{min}}_{\mathrm{TES}} \\
&\text{(IV)} && \dot{Q}^{\mathrm{in}}_{\mathrm{TES},k} \leq \delta^{\mathrm{in}}_{\mathrm{TES},k} \, \frac{E^{\mathrm{nom}}_{\mathrm{TES}}}{\tau^{\mathrm{in}}} \\
&\text{(V)} && -\dot{Q}^{\mathrm{in}}_{\mathrm{TES},k} \leq -\delta^{\mathrm{in}}_{\mathrm{TES},k} \, \dot{Q}^{\mathrm{in,min}}_{\mathrm{TES}} \\
&\text{(VI)} && \dot{Q}^{\mathrm{out}}_{\mathrm{TES},k} \leq \delta^{\mathrm{out}}_{\mathrm{TES},k} \, \frac{E^{\mathrm{nom}}_{\mathrm{TES}}}{\tau^{\mathrm{out}}} \\
&\text{(VII)} && -\dot{Q}^{\mathrm{out}}_{\mathrm{TES},k} \leq -\delta^{\mathrm{out}}_{\mathrm{TES},k} \, \dot{Q}^{\mathrm{out,min}}_{\mathrm{TES}} \\
&\text{(VIII)} && E_{\mathrm{TES}}(t=0) = E_{\mathrm{TES}}(t=t_f) \qquad \text{Cycle constraint} \\
&\text{(IX)} && \delta^{\mathrm{in}}_{\mathrm{TES},k} + \delta^{\mathrm{out}}_{\mathrm{TES},k} \leq 1
\end{aligned}$$

$\forall t \in [0, t_f], \forall k = 1, \ldots, N$. Note that the most likely case is that the TES can be discharged fully and that there are no minimum charging or discharging fluxes, then $E^{\mathrm{min}}_{\mathrm{TES}} = \dot{Q}^{\mathrm{in,min}}_{\mathrm{TES}} = \dot{Q}^{\mathrm{out,min}}_{\mathrm{TES}} = 0$.

Boiler B_i

$$\begin{aligned}
&\text{(I)} && \dot{Q}^{\mathrm{out}}_{\mathrm{B},i,k} = \dot{Q}^{\mathrm{out,nom}}_{\mathrm{B}} \left[\delta_{\mathrm{B},i,k} \, \lambda^{\mathrm{out,min}}_{\mathrm{B}} + \frac{1}{\beta_B} \left(\frac{\dot{Q}^{\mathrm{in}}_{\mathrm{B},i,k} \, \eta^{\mathrm{nom}}_{\mathrm{B}}}{\dot{Q}^{\mathrm{out,nom}}_{\mathrm{B}}} - \delta_{\mathrm{B},i,k} \, \lambda^{\mathrm{in,min}}_{\mathrm{B}} \right) \right] \\
&\text{(II)} && \dot{Q}^{\mathrm{in}}_{\mathrm{B},i,k} \leq \delta_{\mathrm{B},i,k} \, \frac{\dot{Q}^{\mathrm{out,nom}}_{\mathrm{B}}}{\eta^{\mathrm{nom}}_{\mathrm{B}}} \\
&\text{(III)} && -\dot{Q}^{\mathrm{in}}_{\mathrm{B},i,k} \leq -\delta_{\mathrm{B},i,k} \, \lambda^{\mathrm{in,min}}_{\mathrm{B}} \, \frac{\dot{Q}^{\mathrm{out,nom}}_{\mathrm{B}}}{\eta^{\mathrm{nom}}_{\mathrm{B}}}
\end{aligned}$$

$\forall k = 1, \ldots, N, \quad \forall i \in \mathcal{B} = \{1, 2\}$

CHP$_i$

$$\text{(I)}\quad \dot{Q}^{\text{out}}_{\text{CHP},i,k} = \dot{Q}^{\text{out,nom}}_{\text{CHP}} \left[\delta_{\text{CHP},i,k}\, \lambda^{\text{out,min}}_{\text{CHP,th}} + \frac{1}{\beta_{\text{CHP}_{\text{th}}}} \left(\frac{\dot{Q}^{\text{in}}_{\text{CHP},i,k}\, \eta^{\text{nom}}_{\text{CHP,th}}}{\dot{Q}^{\text{out,nom}}_{\text{CHP}}} - \delta_{\text{CHP},i,k}\, \lambda^{\text{in,min}}_{\text{CHP}} \right) \right]$$

$$\text{(II)}\quad P^{\text{out}}_{\text{CHP},i,k} = P^{\text{out,nom}}_{\text{CHP}} \left[\delta_{\text{CHP},i,k}\, \lambda^{\text{out,min}}_{\text{CHP,el}} + \frac{1}{\beta_{\text{CHP}_{\text{el}}}} \left(\frac{\dot{Q}^{\text{in}}_{\text{CHP},i,k}\, \eta^{\text{nom}}_{\text{CHP,el}}}{P^{\text{out,nom}}_{\text{CHP}}} - \delta_{\text{CHP},i,k}\, \lambda^{\text{in,min}}_{\text{CHP}} \right) \right]$$

$$\text{(III)}\quad \dot{Q}^{\text{in}}_{\text{CHP},i,k} \leq \delta_{\text{CHP},i,k} \frac{\dot{Q}^{\text{out,nom}}_{\text{CHP}}}{\eta^{\text{nom}}_{\text{CHP,th}}}$$

$$\text{(IV)}\quad -\dot{Q}^{\text{in}}_{\text{CHP},i,k} \leq -\delta_{\text{CHP},i,k}\, \lambda^{\text{in,min}}_{\text{CHP}} \frac{P^{\text{out,nom}}_{\text{CHP}}}{\eta^{\text{nom}}_{\text{CHP,el}}}$$

$\forall k = 1, \ldots, N, \quad \forall i \in \mathcal{CHP} = \{1, 2\}$

Demand Note that the TES being continuous time but the demand only being given in discrete steps, the actual heat delivered by the TES is first integrated and then averaged.

$$\text{(I)}\quad \dot{Q}_{\text{D},k} = \sum_{i \in \mathcal{CHP}} \dot{Q}^{\text{out}}_{\text{CHP},i,k} + \sum_{i \in \mathcal{B}} \dot{Q}^{\text{out}}_{\text{B},i,k} + \dot{Q}^{\text{out}}_{\text{TES},k} - \dot{Q}^{\text{in}}_{\text{TES},k}$$

$$\text{(II)}\quad P_{\text{D},k} = \sum_{i \in \mathcal{CHP}} P^{\text{out}}_{\text{CHP},i,k} + P_{\text{grid},k}$$

$\forall k = 1, \ldots, N$

Parameters β is the slope of the part-load efficiency curve for the boilers and the electrical and thermal outputs of the CHP respectively.

$$\beta_j = \frac{1 - \lambda^{\text{in,min}}_j}{1 - \lambda^{\text{out,min}}_j}, \qquad j \in \{\text{B, CHP}_{\text{th}}, \text{CHP}_{\text{el}}\}$$

Other parameters have the pre-defined numeric values shown below.

General

$$
\begin{aligned}
\Delta t &= 1\,\mathrm{h} \\
t_f &= 168\,\mathrm{h} \quad \text{(7 days)}
\end{aligned}
$$

$c_{\mathrm{G}}, c_{\mathrm{el}}$ uncertain parameters sampled within ranges $[R_{\mathrm{G}}^{+}, R_{\mathrm{G}}^{-}]$ and $[R_{\mathrm{el}}^{+}, R_{\mathrm{el}}^{-}]$

TES

$$
\begin{aligned}
\tau^{\mathrm{loss}} &= 200\,\mathrm{h} \\
\tau^{\mathrm{in}} &= \tau^{\mathrm{out}} = 1\,\mathrm{h} \\
\eta_{\mathrm{TES}}^{\mathrm{in}} &= \eta_{\mathrm{TES}}^{\mathrm{out}} = 0.95 \\
a &\approx 0.995012 \\
b_1 &\approx 0.94763\,\mathrm{h} \\
b_2 &\approx -1.05\,\mathrm{h} \\
E_{\mathrm{TES}}^{\mathrm{nom}} &= 1000\,\mathrm{kWh} \\
E_{\mathrm{TES}}^{\mathrm{min}} &= \dot{Q}_{\mathrm{TES}}^{\mathrm{in,min}} = \dot{Q}_{\mathrm{TES}}^{\mathrm{out,min}} = 0
\end{aligned}
$$

Boiler

$$
\begin{aligned}
\dot{Q}_{\mathrm{B}}^{\mathrm{out,nom}} &= 530\,\mathrm{kW} \\
\eta_{\mathrm{B}}^{\mathrm{nom}} &= 0.8 \\
\lambda_{\mathrm{B}}^{\mathrm{in,min}} &= 0.173 \\
\lambda_{\mathrm{B}}^{\mathrm{out,min}} &= 0.2
\end{aligned}
$$

CHP

$$
\begin{aligned}
\dot{Q}_{\mathrm{CHP}}^{\mathrm{out,nom}} &= 470\,\mathrm{kW} \\
P_{\mathrm{CHP}}^{\mathrm{out,nom}} &= 380\,\mathrm{kW} \\
\eta_{\mathrm{CHP,th}}^{\mathrm{nom}} &= 0.481 \\
\eta_{\mathrm{CHP,el}}^{\mathrm{nom}} &= 0.389 \\
\lambda_{\mathrm{CHP,th}}^{\mathrm{in,min}} &= \lambda_{\mathrm{CHP,el}}^{\mathrm{in,min}} = 0.582 \\
\lambda_{\mathrm{CHP,th}}^{\mathrm{out,min}} &= 0.622 \\
\lambda_{\mathrm{CHP,el}}^{\mathrm{out,min}} &= 0.5
\end{aligned}
$$

Demand

$\dot{Q}_{\mathrm{D,k}}, P_{\mathrm{D,k}}$ taken from demands time series

1.1 Discretization of TES Equations

The heat balance equation of the TES **TES**(I) is a **linear** first-order ordinary differential equation. Because of its linearity, it can be written into the state space formulation as follows:

$$
\begin{aligned}
\dot{\mathbf{x}}(t) &= A\,\mathbf{x}(t) + B\,\mathbf{u}(t) \\
\dot{\mathbf{x}}(t) &= \dot{E}_{\mathrm{TES}}(t) \\
\mathbf{x}(t) &= E_{\mathrm{TES}}(t) \\
\mathbf{u}(t) &= \begin{bmatrix} \dot{Q}^{\mathrm{in}}_{\mathrm{TES}}(t) & \dot{Q}^{\mathrm{out}}_{\mathrm{TES}}(t) \end{bmatrix}^\top \\
A &= -\frac{1}{\tau^{\mathrm{loss}}} \\
B &= \begin{bmatrix} \eta^{\mathrm{in}}_{\mathrm{TES}} & -\frac{1}{\eta^{\mathrm{out}}_{\mathrm{TES}}} \end{bmatrix}
\end{aligned}
$$

Linear state space models can be **exactly** discretized using the matrix exponential equations, assuming that the input is discretized with zero-order hold. This is a valid assumption, as we want to define the *decision variables* $\mathbf{u}(t)$ anyways ideally only for every hour.

The theoretical discretization of the linear state space equations is as follows:

$$
\begin{aligned}
\mathbf{x}_{k+1} &= A_d\,\mathbf{x}_k + B_d\,\mathbf{u}_k \\
A_d &= e^{A\Delta t} \\
B_d &= B \cdot \int_0^{\Delta t} e^{A\tau} d\tau
\end{aligned}
$$

Evaluating these equations, the following values are obtained:

$$
\begin{aligned}
A_d &\approx 0.99501 \\
B_d &\approx [0.94763, -1.05]
\end{aligned}
$$

Comparing this with explicit Euler discretization which would use $A_d^E = I_{1\times 1} + A\Delta t = 0.995$ and $B_d^E = B\,\Delta t \approx [0.95, -1.0526]$, the relative errors (per component) are: $\frac{A_d - A_d^E}{A_d} \approx 1.2542{\cdot}10^{-5}$ and $\frac{B_d - B_d^E}{B_d} \approx [-0.0025021, -0.0025021]$. The errors are small due to the large time scale differences between the discharge of the TES and the discretization time-step.

If one were to increase the discretization time-step to e.g. $\Delta t = 10\,\mathrm{h}$, the errors would increase to $\frac{A_d - A_d^E}{A_d} \approx 0.0012925$ and $\frac{B_d - B_d^E}{B_d} \approx [-0.025208, -0.025208]$. The increase in the relative error is of multiple orders of magnitude for A_d^E and one order of magnitude for B_d^E.

The simulated homogeneous response of the TES upon being fully charged is shown in Fig. 1 for both methods and both time-steps.

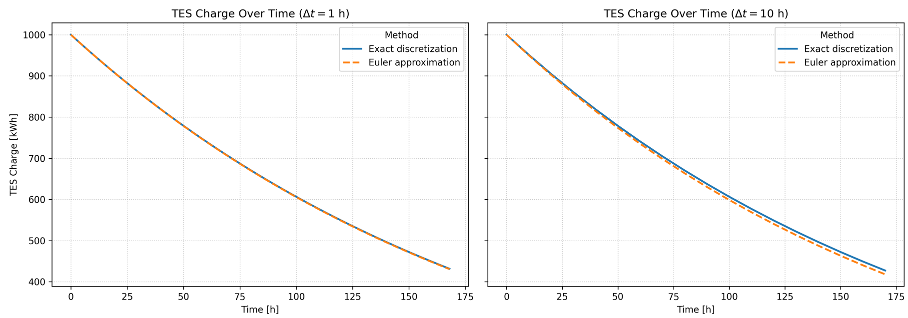

Figure 1: Comparison of discretization methods for the TES differential equation. Plots show the homogeneous response of the TES upon fully charged with $E_{\text{TES}}(t_0) = 1000\,\text{kWh}$ at the beginning.

2 LP Formulation

Most things stay the same between the MILP and the LP. The binary variables are removed from the decision variables, therefore the constraints containing binary variables also change.

Decision variables At each time step $k \in \{1, \ldots, N\}$, the decision variable vector of the independent variables is

$$\tilde{\mathbf{x}}_k = \left[\dot{Q}^{\text{in}}_{\text{B},1,k},\ \dot{Q}^{\text{in}}_{\text{B},2,k},\ \dot{Q}^{\text{in}}_{\text{CHP},1,k},\ \dot{Q}^{\text{in}}_{\text{CHP},2,k},\ \dot{Q}^{\text{in}}_{\text{TES},k},\ \dot{Q}^{\text{out}}_{\text{TES},k},\ P_{\text{grid},k}\right]$$

With all variables being continuous and in $\mathbb{R}^n$. The auxiliary variables stay the same.

2.1 Linearization of constraints

The part-load behavior of the components contains binary variables. These should be eliminated to get a linear formulation. This has the downside of removing the minimum operating point.

The linearization is obtained by fitting a linear function through the original piece-wise linear efficiency curve. This is done numerically through optimization in `scipy`. This is the minimization problem that is solved:

$$\min_{\alpha_i} \int \left(\alpha_i\, x_i - f_i(x_i)\right)^2 dx_i$$

$$\text{s.t.} \quad f_i(x_i) = \begin{cases} x_i^{\text{out,nom}} \left[\lambda_i^{\text{out,min}} + \frac{1}{\beta_i}\left(\frac{x_{i,k}^{\text{in}}\, \eta_i^{\text{nom}}}{x_i^{\text{out,nom}}} - \lambda_i^{\text{in,min}}\right)\right] & \text{if } x_{i,k}^{\text{in}} \geq x_{i,k}^{\text{in,min}} \\ 0 & \text{otherwise} \end{cases}$$

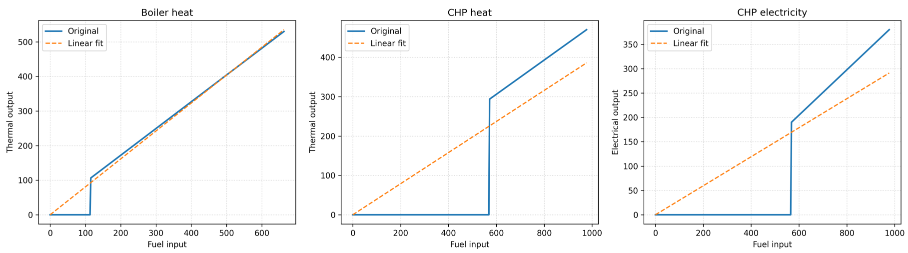

Figure 2: Linearization results for the 3 components.

The objective is equivalent to minimizing the mean squared error between the two functions. This optimization results in the following constants:

$$\alpha_{\mathrm{B}} = 0.8078 \tag{1}$$
$$\alpha_{\mathrm{CHP,th}} = 0.3948 \tag{2}$$
$$\alpha_{\mathrm{CHP,el}} = 0.298 \tag{3}$$

See Fig. 2 for the linearization results.